

PROJECT REPORT

Image Recognition Using Pre-trained MobileNetV2

Submitted By

Name: Huzaifa Ijaz

Batch: 2026

Organization: DecodeLabs

Course: Artificial Intelligence

Project: Project 4 – Image Recognition

Abstract

Image recognition is one of the most widely used applications of Artificial Intelligence and Computer Vision. It enables computers to identify and classify objects present in digital images. Modern image recognition systems use deep learning models trained on millions of images to recognize thousands of different object categories.

In this project, a pre-trained MobileNetV2 model provided by TensorFlow was used to perform image classification. The model predicts the most likely object present in a user-provided image along with confidence scores. This project demonstrates the practical use of transfer learning and pre-trained neural networks without requiring model training from scratch.

1. Introduction

Artificial Intelligence (AI) has significantly improved the ability of computers to understand visual information. Image recognition allows machines to identify objects, people, animals, and scenes automatically.

Deep learning models such as MobileNetV2 are trained on very large datasets and can recognize thousands of object categories with high accuracy.

This project implements a simple image recognition system using TensorFlow and a pre-trained MobileNetV2 model.

2. Objectives

The objectives of this project are:

- Understand image recognition using Artificial Intelligence.
 - Learn how pre-trained deep learning models work.
 - Load and preprocess an input image.
 - Perform image classification.
 - Display prediction results with confidence scores.
 - Explore practical applications of computer vision.
-

3. Problem Statement

Manual identification of objects from images is time-consuming and not scalable for large datasets. AI-based image recognition automates this process by classifying images quickly and accurately.

The objective of this project is to build a basic image recognition system capable of recognizing objects from an input image using a pre-trained deep learning model.

4. Software Requirements

Programming Language

- Python 3.x

Libraries

- TensorFlow
- NumPy
- Pillow

Development Environment

- Visual Studio Code
- Jupyter Notebook (Optional)

5. Hardware Requirements

- Windows/Linux/macOS
- Intel i3 or higher processor
- 4 GB RAM (Minimum)
- Internet connection (required once to download the pre-trained model)

6. Technology Used

TensorFlow

TensorFlow is an open-source machine learning framework developed by Google. It provides tools for building, training, and deploying deep learning models.

MobileNetV2

MobileNetV2 is a lightweight convolutional neural network optimized for image classification tasks. It is trained on the ImageNet dataset containing over one million images across 1,000 object categories.

7. Methodology

The project follows the following workflow:

1. Load the MobileNetV2 pre-trained model.
 2. Ask the user to provide an image.
 3. Resize the image to 224×224 pixels.
 4. Convert the image into numerical arrays.
 5. Preprocess the image.
 6. Feed the image into the neural network.
 7. Predict object classes.
 8. Display the Top-5 predictions with confidence scores.
-

8. System Flow

Input Image



Image Preprocessing



MobileNetV2 Model



Prediction



Top 5 Results



Output Display

9. Implementation

The implementation consists of the following steps:

- Import required Python libraries.
- Load the MobileNetV2 pre-trained model.
- Read the image from the user.
- Resize and preprocess the image.
- Perform prediction.
- Decode prediction labels.
- Display the results.

10. Results

The developed system successfully identifies objects from images.

Example Output

Input Image:

cat.jpg

Predictions:

Rank	Prediction	Confidence
1	Tabby Cat	97.83%

Rank	Prediction	Confidence
2	Tiger Cat	1.46%
3	Egyptian Cat	0.41%
4	Lynx	0.17%
5	Persian Cat	0.08%

The model correctly identifies the object with high confidence.

11. Advantages

- High prediction accuracy.
 - No training required.
 - Fast image classification.
 - Easy to implement.
 - Lightweight architecture.
 - Suitable for real-world applications.
-

12. Limitations

- Limited to ImageNet object categories.
 - Requires internet access the first time the model is downloaded.
 - Performance depends on image quality.
 - Cannot recognize custom objects without additional training.
-

13. Applications

Image recognition is widely used in:

- Face Recognition
- Medical Imaging

- Autonomous Vehicles
 - Smart Surveillance
 - Agriculture
 - Wildlife Monitoring
 - Retail Automation
 - Manufacturing
 - Robotics
 - Security Systems
-

14. Future Enhancements

The project can be extended by:

- Building a real-time webcam recognition system.
 - Detecting multiple objects in one image.
 - Training custom datasets.
 - Deploying as a web application using Flask or Streamlit.
 - Integrating object detection models such as YOLO.
-

15. Conclusion

This project successfully demonstrates the implementation of an AI-based image recognition system using TensorFlow's MobileNetV2 model. The model accurately classifies objects in images without requiring additional training.

The project provides practical experience with deep learning, transfer learning, image preprocessing, and computer vision. It serves as an excellent introduction to real-world AI applications and lays the foundation for more advanced image recognition and object detection projects.

16. Learning Outcomes

After completing this project, the following concepts were learned:

- Artificial Intelligence fundamentals
 - Deep Learning basics
 - Computer Vision
 - Transfer Learning
 - Image Preprocessing
 - TensorFlow Framework
 - MobileNetV2 Architecture
 - Prediction using Pre-trained Models
-

17. References

1. TensorFlow Documentation
 2. Keras Applications Documentation
 3. ImageNet Dataset Documentation
 4. Python Documentation
 5. DecodeLabs Artificial Intelligence Internship Project Guide
-

Appendix

Project Files

- project4.py
- requirements.txt
- README.md
- Project4_Report.pdf
- Sample Images
- Output Screenshots

Required Libraries

- tensorflow

- numpy
- pillow

Folder Structure

AI-Project-4-Image-Recognition/

```
|  
├── project4.py  
├── requirements.txt  
├── README.md  
├── Project4_Report.pdf  
├── sample.jpg  
└── Screenshots/
```